

Question Number	Answer	Additional Guidance	Mark
1ai	An answer that includes the following points • (using a light microscope find cell) under low power (1) • then view under high power (1) • calibrate eyepiece graticule / use of stage micrometer • count number of (eyepiece) graticule units over the cell (1) • convert eyepiece graticule units (to microns using calibration) (1) • measure diameter at different positions / orientations (1)	IGNORE electron microscope	expert 5

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1aii	A drawing with the following features • drawing using continuous lines showing correct shape of cell (1) • nucleus drawn with continuous lines in correct position and correct size (1) • any two correct labels (1) • suitable scale line shown (1)	Lose this mark if draw cell wall membrane / cytoplasm / nucleus / mitochondria IGNORE cell wall, wrong labels as long as 2 are correct could be a scale line with suitable measurement / magnification (about 1000 if a 6 cm drawing, others pro-rata)	4

Question Number	Answer	Additional Guidance	Mark
1aiii	A calculation which includes the following steps • conversion of onion cell length to micrometres / cheek cell to mm (1) • division of onion cell length by diameter of human cell and answer quoted as a ratio (1)	= 200 (μm) OR 0.06 mm = 200 $\div$ 60 OR 0.2 $\div$ 0.06 3.33 : 1 ACCEPT 3.3, 3.33 ACCEPT 1:0.3 NOT 1:3.33, this gets 1 mark	2

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1bi	The following correctly named: • X- Sclerenchyma, Y- Phloem / Sieve (tube element), Z- Xylem (cell / vessel) (2) All three for 2 marks 1 or 2 for one mark.		2

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1bii	• {phloem / sieve} cell / correct letter from 1bi	consider ecf from bi	1